Banking ETL Pipeline

Analytics Dashboard

End-to-end data pipeline processing 25,500 banking transactions with automated quality checks, trend analysis & risk intelligence

25,500	98.4%	5	3
Records Processed	Load Success Rate	SQL Analysis Views	Dashboard Pages

Consumer & Community Banking · Data Analytics & Reporting · 2024

Agenda

What this deck covers

01 **Business Context & Problem Statement**
Why this pipeline was built and what business question it answers

02 **Solution Architecture**
ETL design — Extract, Transform, Load — and technology choices

03 **Data Quality & Pipeline Health**
How 25,500 records were validated, cleaned and tracked

04 **Executive KPIs & Trend Analysis**
Monthly volume, success rates and channel performance

05 **Customer Intelligence**
Segmentation — Platinum, Gold, Silver, Standard — and spend patterns

06 **Risk & Fraud Intelligence**
Failure rate analysis and high-risk customer flagging

07 **Key Findings & Recommendations**
Actionable insights for the business

Business Context

The problem this pipeline was designed to solve

The Challenge

Consumer banking teams generate 25,000+ transactions monthly across 5 channels. Without a structured pipeline, analysts spent hours manually cleaning data — leaving no time for the trend analysis and risk flagging that drives business decisions.

PROBLEM

Manual & Error-Prone

- Data scattered across raw CSVs
- No validation or quality checks
- Duplicate records went undetected
- No audit trail of processing

SOLUTION

Automated ETL Pipeline

- Python extracts & stages raw data
- 7-step transform & validation
- MySQL production schema
- Full audit log per run

IMPACT

Business Value Delivered

- 25,103 clean records loaded
- Risk customers auto-flagged
- Monthly trends auto-reported
- Zero manual intervention

Solution Architecture

Python + MySQL ETL pipeline — end-to-end flow

EXTRACT
extract.py

Reads raw CSV
25,500 records
into stg_transactions

VALIDATE
transform.py

7 quality checks
Null, type, status
dedup validation

ENRICH
transform.py

is_large_txn flag
txn_month column
business rules

LOAD
load.py

fact_transactions
production table
upsert logic

ANALYSE
MySQL Views

5 SQL views
trend, risk, segment
channel, category

VISUALISE
Power BI

3-page dashboard
live MySQL connect
slicers + drilldowns

TECHNOLOGY STACK

Python 3.13
Extract, Transform, Load scripting

MySQL 8.0
Staging + production schema

PyMySQL / Pandas
DB connector + data manipulation

Power BI Desktop
Interactive dashboard & reporting

uuid + openpyxl
Data generation + Excel export

Data Quality: 7-Step Transform

1. Strip whitespace
2. Standardise casing
3. Drop null critical fields
4. Validate amounts
5. Validate dates
6. Validate status values
7. Remove duplicates

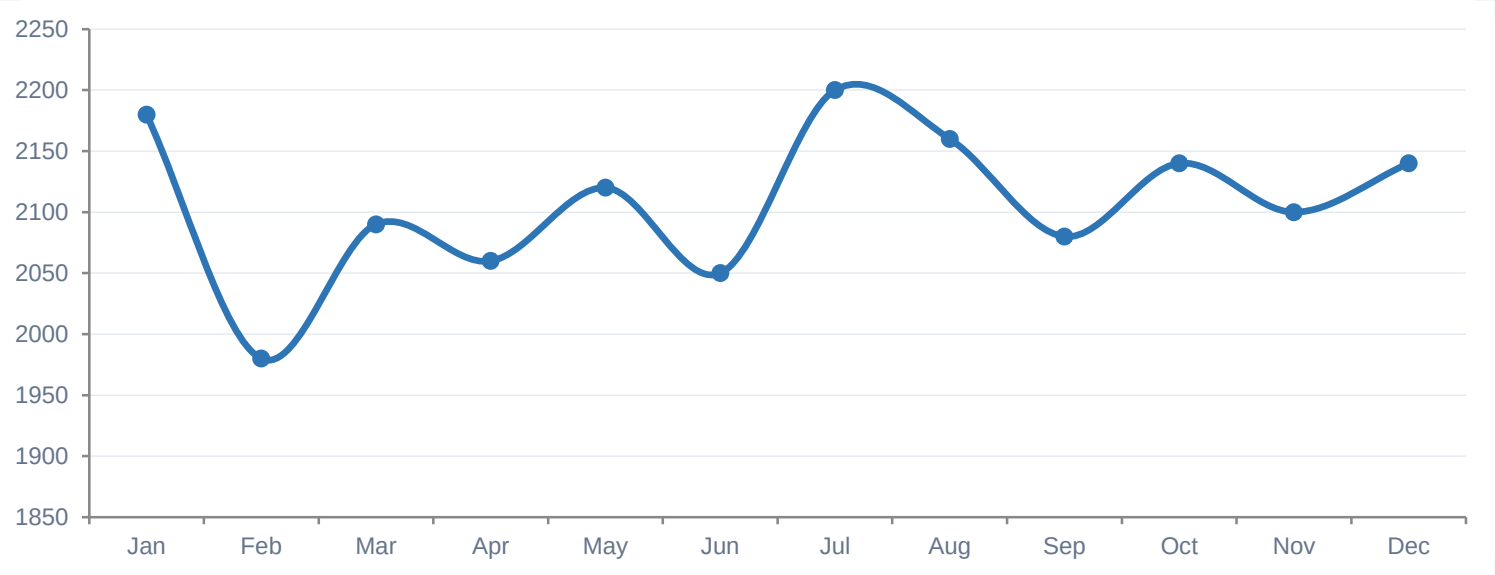
397 records rejected · 25,103 loaded · 98.4% success

Executive KPIs

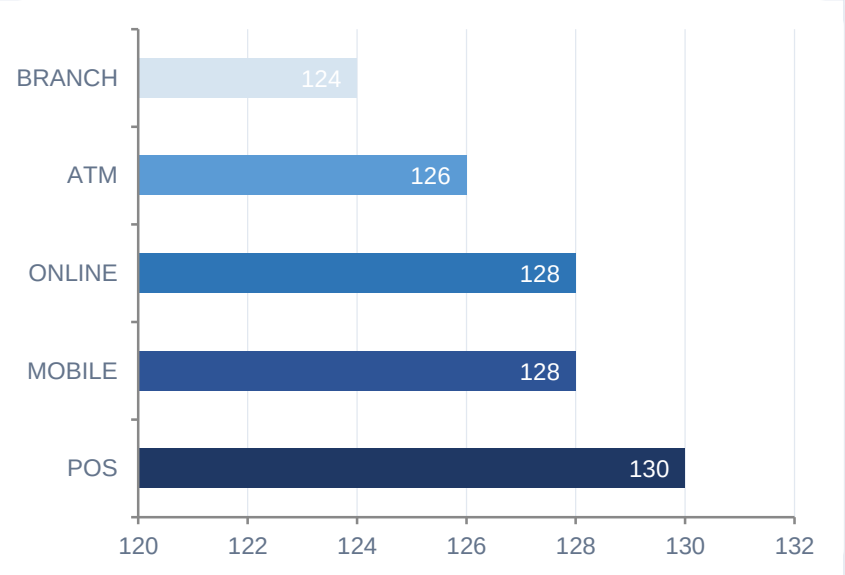
Snapshot of pipeline output and transaction health — FY 2024

25,103	₹634.78M	60.1%	6
Total Transactions	Total Volume	Success Rate	High Risk Customers

MONTHLY TRANSACTION VOLUME



VOLUME BY CHANNEL



Key Insight

Transaction volumes remained stable across all 12 months (avg. 2,100/month) with no seasonal concentration risk. All 5 channels contribute near-equal volume (124–130M), indicating healthy channel diversification with no single point of failure.

Customer Intelligence

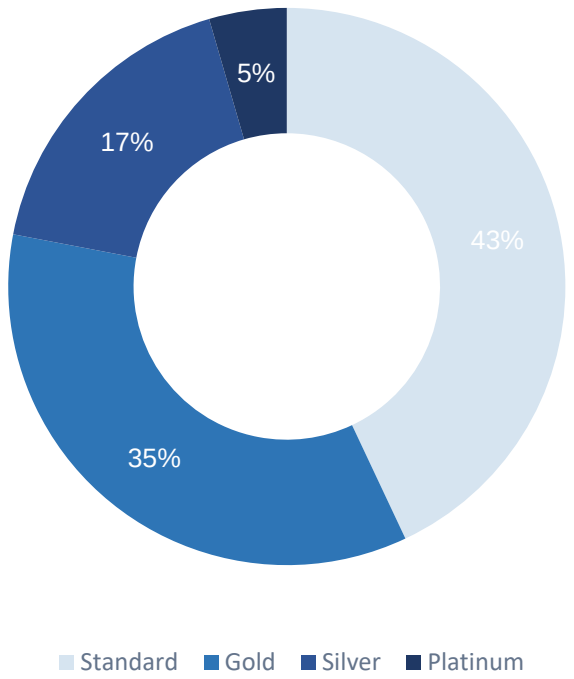
Segmentation and spend analysis across 200 customers

200
Total Customers

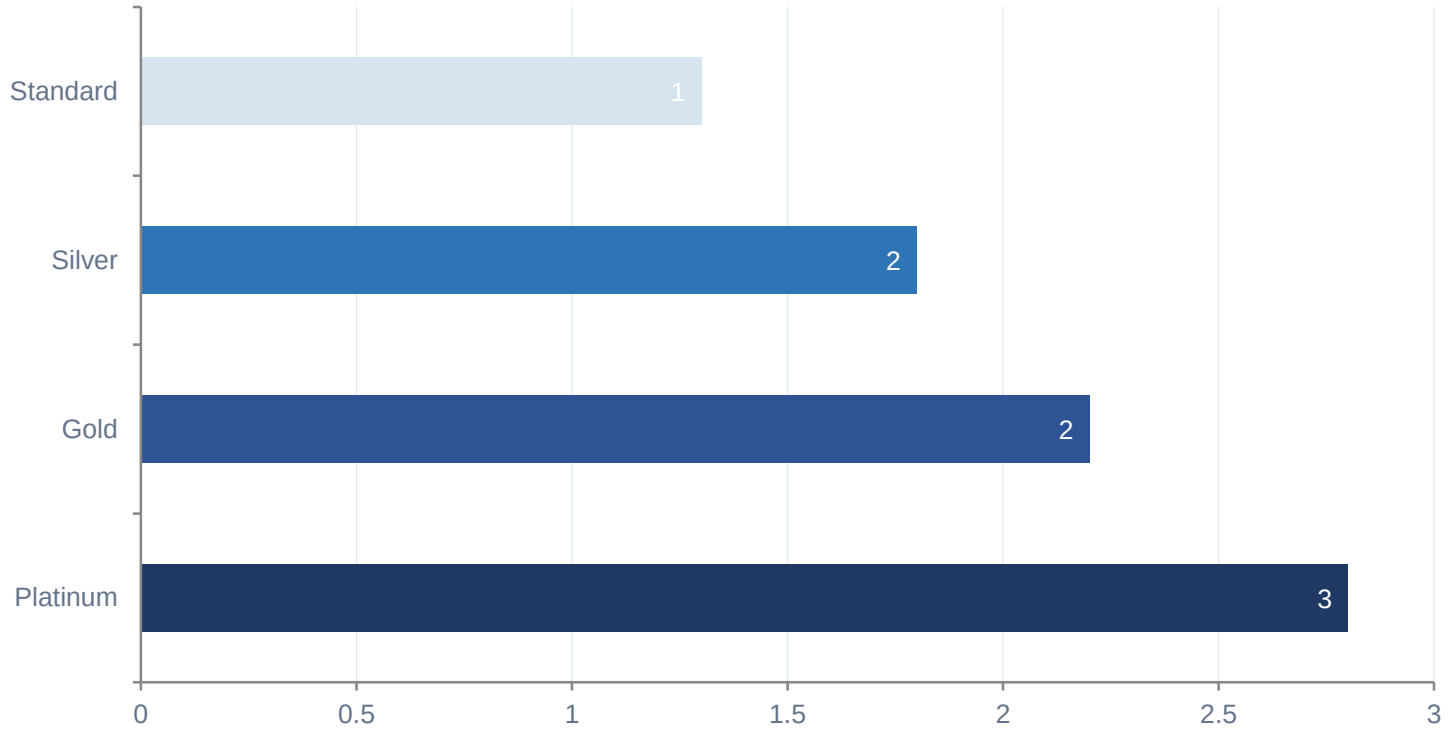
9
Platinum Customers

₹1.88M
Avg Spend / Customer

SEGMENT DISTRIBUTION



AVG SPEND BY SEGMENT



Platinum customers (4.5% of base) generate 2.15x the spend of Standard customers — targeted retention of this segment is high-priority.

Risk & Fraud Intelligence

Failure rate analysis and high-risk customer identification

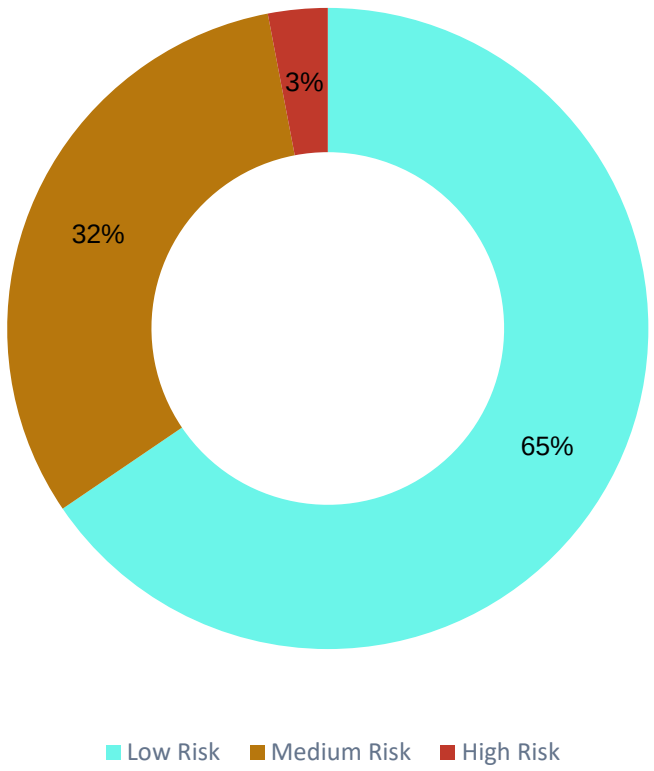
137
Total Flagged Customers

21.9%
Avg Failure Rate

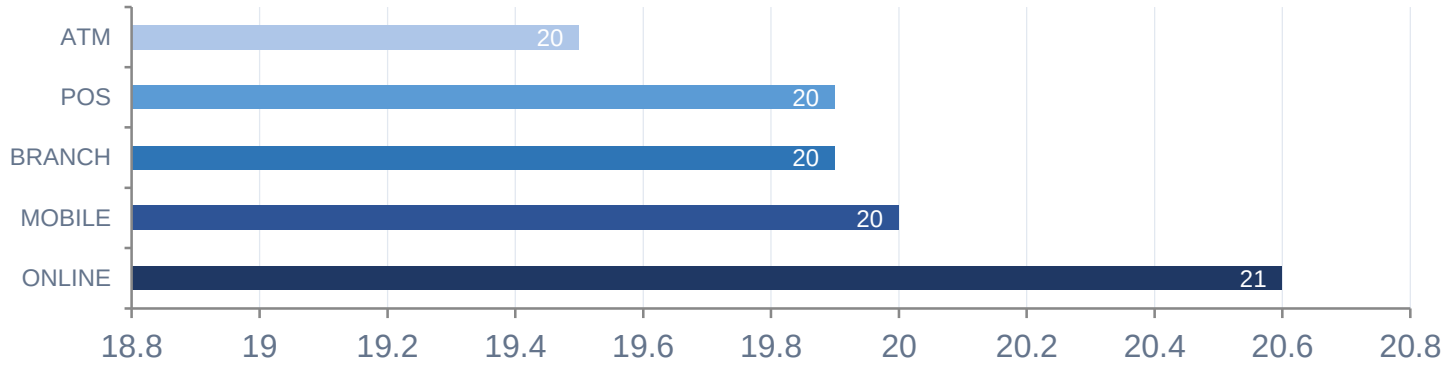
29.6%
High Risk Failure Rate

21.5%
Medium Risk Failure Rate

RISK DISTRIBUTION



FAILURE RATE BY CHANNEL



TOP HIGH RISK CUSTOMERS

Customer ID	Total Txns	Failed	Fail Rate %	Risk Level
CUST0116	130	41	31.54	HIGH RISK
CUST0089	110	33	30.00	HIGH RISK
CUST0045	125	37	29.60	HIGH RISK
CUST0039	129	35	27.13	MEDIUM RISK

Failure rates are near-uniform across channels (19.5–20.6%) suggesting a systemic issue rather than a channel-specific one — recommend reviewing authentication and timeout policies.

Key Findings & Recommendations

Actionable insights for the Consumer & Community Banking business

01 Channel Diversification is Strong

Finding:

All 5 channels (POS, Mobile, Online, ATM, Branch) contribute near-equal volume (124–130M), eliminating single-channel dependency.

Recommendation:

Invest equally across channels. No consolidation risk identified.

02 Platinum Segment Drives Outsized Value

Finding:

9 Platinum customers (4.5% of base) generate 2.15x the spend of Standard customers, with avg spend of ₹2.8M per customer.

Recommendation:

Launch dedicated Platinum retention programme with personalised relationship managers.

03 Systemic Failure Rate Requires Attention

Finding:

Failure rates are near-uniform across all channels (19.5–20.6%), pointing to a systemic backend issue rather than channel-specific problems.

Recommendation:

Audit authentication, timeout, and retry policies. Target <10% failure rate within Q2.

04 ETL Pipeline Delivers Reliable Data

Finding:

98.4% of records loaded successfully. 397 records rejected with full audit trail. Zero manual intervention required.

Recommendation:

Schedule daily automated runs. Integrate rejection alerts into ops monitoring.

Thank You

Built with Python · MySQL · Power BI

25,500	7	5	3	98.4%
Records Processed	Quality Checks	SQL Views	Dashboard Pages	Load Success